

PROTEOMICS

Towards resolving proteomes in single cells

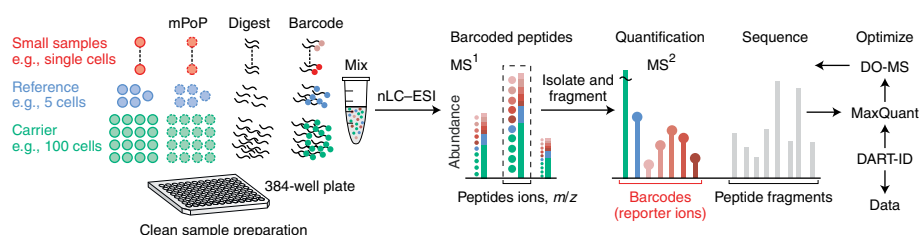
Development of methods for single-cell proteomics increases depth and throughput

Complex systems such as tissues and organs are made up of functionally distinct cells. This difference stems from a difference in the proteomic content of individual cells, and characterizing this cellular heterogeneity is crucial to understanding cellular differentiation and development, classifying cancers, diagnosing diseases and discovering biomarkers. However, bulk proteome measurements provide population averages, and determining the contributions of different cell types is not straightforward.

Single-cell RNA sequencing approaches provide an indirect readout for protein levels in a given cell, but these measurements tend to be sparse for many transcripts. Additionally, protein stability influences the proteome, and this information cannot be derived from RNA-seq, making it necessary to directly quantify the proteome in single cells. Antibody-based methods have been employed to quantify proteomes in single cells, but these have limited specificity. Recently, liquid chromatography–mass spectrometry (LC–MS)-based proteomics approaches have made headway in increasing the throughput and depth of single-cell measurements.

Nikolai Slavov and his research group at Northeastern University previously developed SCoPE-MS (Single-Cell Proteomics by Mass Spectrometry)—a single-cell proteomics method that originally made use of carrier cells with isobaric labeled proteins mixed with single cells of interest to reduce sample loss. The group has now published SCoPE2, which includes improvements to sample preparation, data acquisition, data analysis and integration across different TMT batches, increasing in the number of cells and proteins analyzed (Specht et al. 2021). “SCoPE2 uses only standard commercial equipment for automated single-cell MS analysis of about 200 single cells per 24 hours. Using SCoPE2, we could quantify over 3,000 proteins across multiple single cells,” says Slavov.

The team used SCoPE2 to look at the differentiation of homogeneous monocytes into macrophage-like cells. “The data revealed a continuous gradient of proteome states for macrophages differentiated from monocytes. The gradient strongly correlated



Workflow of the SCoPE2 method. mPoP, Minimal Proteomic sample Preparation; DO-MS, Data-driven Optimization of MS; DART-ID, Data-driven Alignment of Retention Times for Identification; nLC-ESI, nano-scale liquid chromatography and electrospray ionization MS. Credit: Adapted from H. Specht. et al. *Genome Biol.* 22, 50, 2021, under a [Creative Commons Attribution 4.0 International License](https://creativecommons.org/licenses/by/4.0/)

to the inflammatory axis of classically and alternatively activated macrophages, suggesting the emergence of polarization in the absence of cytokines,” says Slavov, highlighting the power of single-cell proteomics to facilitate new biological discoveries. “A significant novelty of the paper is the joint analysis of proteomes and transcriptomes of single cells from the same population,” he adds.

At the Technical University of Denmark and University of Copenhagen, Erwin Schoof and colleagues have developed a single-cell proteomics workflow also inspired by SCoPE-MS (Schoof et al. 2021). This method retains as well the FACS parameters for each cell from the sorting step; these are later overlaid with the proteomics data to annotate cell identity, allowing the researchers to perform single-cell proteomics measurements on a complex mixture. Until now, “single-cell proteomics papers have often used homogeneous cell lines as their model systems and sorted out predefined cell populations, measured them separately and then demonstrated the ability to tell them apart by their expressed proteomes. We for the first time sorted single cells from a heterogeneous ex vivo leukemia model system in bulk and demonstrate the ability to detect the three cell types we functionally know are contained therein (leukemia stem cells, progenitors and blasts),” explains Schoof. They were also able to detect and functionally validate an alternative differentiation trajectory of leukemia stem cells into blasts that does not pass through

the conventional progenitor stage. “This would have never been detected using a classical bulk proteomics approach,” says Schoof.

Single-cell proteomics methods still leave much to be desired. Both approaches currently analyze about 200 single cells per day and about 1,000 proteins per cell. A challenge that still remains is that of missing values and coverage. “While we are able to consistently read out more than 1,000 proteins per cell, we would like to go towards 3,000–5,000 proteins per cell to get a better representation of which cellular proteomes give rise to which cellular phenotypes,” says Schoof. Both groups think that better sample preparation methods, alternative experimental designs and instrumentation, as well as new data analysis approaches, could alleviate this limitation. Slavov hopes to develop the approach in the direction of multimodal analysis in which RNA, metabolites, proteins and their post-translational modifications could be simultaneously analyzed from each single cell.

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Research papers

Specht, H. et al. Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity using SCoPE2. *Genome Biol.* 22, 50 (2021).
 Schoof, E. M. et al. Quantitative single-cell proteomics as a tool to characterize cellular hierarchies. *Nat. Commun.* 12, 3341 (2021).